W-H-193D
September 13, 1977
SUPERSEDING
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December 30, 1974

FEDERAL SPECIFICATION

HEATER, SPACE, ELECTRIC (PORTABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable electric space heaters (see 6.1).

1.2 Classification. Portable space heaters shall be of the following types and classes as specified (see 6.2):

Type I - Upright, Convection, Fan Forced, 120 volts

Class A - Dual wattage: 1300 watts (± 50 watts) and 1500 watts
Class B - Single wattage: 1250 watts (min.)

Type II - Upright, Radiant, Fan Forced, 120 volts

Class A - Dual wattage: 1300 watts (± 50 watts) and 1500 watts
Class B - Single wattage: 1250 watts (min.)

Type III - Low Silhouette, Natural Convection, 120 volts

Class A - Triple wattage: 500 watts, 1000 watts and 1500 watts
Class B - Single wattage: 1000 watts (min.)

Type IV - Low Silhouette, Radiant, Fan Forced, 120 volts

Class A - Dual wattage: 1300 watts (± 50 watts) and 1500 watts
Class B - Single wattage: 1250 watts (min.)

Type V - Upright, Convection, Fan Forced, 240 volts

Class A - Dual wattage: 2000 watts and 4000 watts
Class B - Single wattage: 3000 watts

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard; Corrugated, Triple-Wall.

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters Labortories, Inc., (UL) Standards:

UL 1025 - Electric Air Heaters.
UL 1042 - Electric Baseboard Heating Equipment.

(Application for copies should be addressed to the nearest Underwriters' Laboratories, Inc., at 207 East Ohio Street, Chicago, IL 60611; Walt Whitman Road, Melville, Long Island, NY 11759; 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Standard product. The portable heaters provided under this specification shall be the manufacturer's standard commercial product, except for any design modification and additional requirements specified herein.

3.2 Fire and casualty hazards. Each contractor shall submit to the contracting agency proof that the heaters he proposes to supply under this specification conform to the requirements of the applicable UL Standard. Listing by Underwriters' Laboratories will be accepted as evidence that the product conforms to this requirement. In lieu of Underwriters' Listing, the contractor may submit independent proof satisfactory to the contracting agency that the heaters conform to applicable requirements of the published standard, including method of tests, of the applicable UL Standard listed in 2.2. Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his material or equipment.

3.3 Workmanship. The finished product shall be clean, well made, and free from defects which may affect appearance, serviceability, or durability.

3.4 Tip-over switch. All heaters, except type V, shall be furnished with a tip-over switch. This switch shall automatically cut off the power supply when the heater is accidentally turned over, forward or backward, from its normal operating position. The switch shall automatically re-activate the power supply when the heater is returned to its normal operating position.

3.5 Electrical cord and plug. Each heater shall be equipped with a cord and plug meeting the applicable UL 1025 or UL 1042 requirements. The length of the cord shall not be less than 6 feet. The color shall be the manufacturer's standard.

3.5.1 One-hundred and twenty volts heaters. Each 120 volts heater shall be equipped with a cord and plug rated for 125 volt, 15 amp., 2 pole, 3 wire grounding.

3.5.2 Two-hundred and forty volts heaters. Each 240 volts heater shall be equipped with a cord and plug rated for 250 volts, 2 pole, 3 wire grounding.

3.5.2.1 For type V, class A heaters, the plug and cord shall be rated for 20 amps.

3.5.2.2 For type V, class B heaters, the plug and cord shall be rated for 15 amps.

3.6 Thermostat control. Each heater shall have an automatic adjustable thermostat with a range of 45° to 120°F or greater. The thermostat shall have a marked "off" or similar marking at which the power supply circuit can be opened. If a "positive off" thermostat is not used, a separate manual "on-off" switch will be acceptable.

3.7 Handle. All heaters shall be equipped with a suitable carrying handle.

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3.8 Type I, upright, convection, fan forced, 120 volts. The elements shall be of the "Black Heat" variety made of chrome-nickle resistance wire. It shall be equipped with a suitable fractional horsepower motor and fan for circulating the heat. The dimensions shall conform to table I. The heater shall operate on 120 volts A.C., 60 HZ. The electrical cord and plug shall be in accordance with 3.5.

3.8.1 Class A. The rated output shall be of a dual range. The lower output shall be rated between 1250 watts and 1350 watts, inclusive. The higher output shall be rated at 1500 watts.

3.8.2 Class B. The rated output shall be of a single wattage of at least 1250 watts.

TABLE I. Type I, upright, convection fan forced heater

Class	Wattage	Height (excluding handle)	Width	Depth
A	1300 (±50) and 1500 watts	17.0 ± 5.0 in.	12.0 ± 4.0 in.	7.5 ± 2.5 in.
B	1250 watts (min.)	15.5 ± 6.5 in.	14.0 ± 5.0 in.	6.0 ± 3.0 in.

3.9 Type II, upright, radiant, fan forced, 120 volts. The elements shall be of the "Instant Heat" variety. Ribbon elements or spiral wire wound elements will be acceptable. It shall be equipped with a suitable fractional horsepower motor and fan for circulating the heat. The dimensions shall conform to table II. The heater shall operate on 120 volts A.C., 60 Hz. The electrical cord and plug shall be in accordance with 3.5.

3.9.1 Class A. The rated output shall be of a dual range. The lower output shall be rated between 1250 watts and 1350 watts, inclusive. The higher output shall be rated at 1500 watts.

3.9.2 Class B. The rated output shall be of a single wattage of at least 1250 watts.

TABLE II. Type II, upright, radiant fan forced heater

Class	Wattage	Height (excluding handle)	Width	Depth
A	1300 (±50) and 1500 watts	13.5 ± 2.0 in.	19.25 ± 5.25 in.	8.50 ± 3.50 in.
B	1250 watts (min.)	12.5 ± 3.5 in.	16.00 ± 5.00 in.	8.75 ± 2.75 in.

3.10 Type III, low silhouette, natural convection, 120 volts. The elements shall be either the rod and fin variety or the rod and helical fin variety. The dimensions shall conform to table III. The heater shall operate on 120 volts A.C., 60 Hz. The electrical cord and plug shall be in accordance with 3.5.

3.10.1 Class A. The rated output shall be of three wattages. The wattages shall be 500, 1000, and 1500 watts.

3.10.2 Class B. The rated output shall be of a single wattage of at least 1000 watts.

TABLE III. Type III, low silhouette, natural convection heater

Class	Wattage	Height (excluding handle)	Width	Depth
A	500, 1000, 1500 watts	6.5 ± 2.5 in.	47.5 ± 12.5 in.	4.25 ± 2.75 in.
B	1000 watts (min.)	6.5 ± 2.5 in.	47.5 ± 12.5 in.	4.25 ± 2.75 in.

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3.11 Type IV, low silhouette, radiant, fan forced, 120 volts. The elements shall be of the "Instant Heat" variety. Ribbon or tubular elements will be acceptable. It shall be equipped with a suitable fractional horsepower motor and fan for circulating the heat. The dimensions shall conform to table IV and to UL 1025, Appendix A. The heater shall operate on 120 volts A.C., 60 Hz. The electrical cord and plug shall be in accordance with 3.5.

3.11.1 Class A. The rated output shall be a dual range. The lower output shall be rated between 1250 watts and 1350 watts, inclusive. The higher output shall be rated at 1500 watts.

3.11.2 Class B. The rated output shall be of a single wattage of at least 1250 watts.

TABLE IV. Type IV, low silhouette, radiant fan forced heater

Class	Wattage	Height (excluding handle)	Width	Depth
A	1300 (+50) and 1500 watts	8.25 ± 0.75 in.	38.5 ± 4.0 in.	3.25 ± 1.75 in.
B	1250 watts (min.)	9.25 ∓ 1.75 in.	36.5 ∓ 6.5 in.	3.25 ∓ 1.75 in.

3.12 Type V, upright, convection, fan forced, 240 volts. The elements shall be of the "Black Heat" variety made of chrome-nickle resistance wire. It shall be equipped with a suitable fractional horsepower motor and fan for circulating the heat. The dimensions shall conform to table V. The heater shall operate on 240 volts A.C. The electrical cord and plug shall be in accordance with 3.5.

3.12.1 Class A. The rated output shall be of a dual range. The lower output shall be rated at 2000 watts. The higher output shall be rated at 4000 watts.

3.12.2 Class B. The rated output shall be of a single wattage of 3000 watts.

TABLE V. Type V, upright, convection, 240 volt fan forced heater

Class	Wattage	Height (excluding handle)	Width	Depth
A	2000 and 4000 watts	17.50 ± 5.50 in.	18.50 ± 3.50 in.	7.0 ± 2.0 in.
B	3000 watts	17.50 ∓ 5.50 in.	18.75 ∓ 3.25 in.	9.0 ∓ 4.0 in.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Dimensional and visual examination. Sampling for these characteristics shall be in accordance with Inspection Level S-3 in MIL-STD-105.

4.2.2 Tests. Sampling for the tests in paragraph 4.4 shall be in accordance with Inspection Level S-1 in MIL-STD-105.

4.3 Inspection.

4.3.1 End item.

4.3.1.1 The heaters shall be examined for defects listed in table VI. The Acceptable Quality Levels (AQL) for defects shall be 2.5% defective for major defects and 6.5% defective for total defects.

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4.3.1.2 The heaters shall be tested in accordance to paragraph 4.4. The Acceptable Quality Levels (AQL) for defects shall be 4.0% defective.

TABLE VI. Classification of defects, end item

Defects	Major	Minor
Type and class, not as specified	X	
Elements not as specified	X	
Thermostat control missing	X	
Handle missing	X	
Tip over safety switch missing when specified	X	
Cord less than 6 feet		X
Dimensions not as specified	X	
Sharp projections		X
Enamel finish rough or blistered		X
Positive off switch missing	X	
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VII. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operation. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE VII. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application or components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.
Contents	Net weight exceeds box specification requirements.

4.4 Test methods.

4.4.1 Portable space heaters. Test portable space heater in accordance with the UL standard listed in table VIII.

TABLE VIII. Test standards

Space heater type	UL Standard for safety (document No.)
I. Upright, convection, fan forced	1025
II. Upright, radiant, fan forced	1025
III. Low silhouette, natural convection	1025 or 1042
IV. Low silhouette, radiant, fan forced	1025 or 1042
V. Upright, convection, 240 volt, fan forced	1025

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each heater shall be packaged in accordance with MIL-P-116, Method IA-14.

5.1.2 Commercial. The heater shall be packaged in unit containers normally used in commercial practice.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A. Heaters of like description packaged as specified in 5.1, shall be packed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant; or to PPP-B-640; class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the wood and triple-walled fiberboard box shall not exceed 200 pounds. The gross weight of PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Commercial. The heaters shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. (Applicable to full rail car and truck load shipments only.) The heaters shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

5.4.1 Civil agencies. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Convection or "Black Heat" type portable heaters (types I and III) are primarily intended to be used for heating small rooms or areas such as guard booths and pump rooms.

6.1.2 Radiant or "Instant Heat" type portable heaters (types II and IV) are primarily intended to be used for spot heating such as an area around a desk.

6.1.3 Two-hundred and forty (240) volts, convection type portable heaters (type V) are primarily intended to be used for heating rooms where more warmth is needed and a 240 volt outlet is available.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class (see 1.2).
(c) Size of lot other than specified (see 4.2.1).
(d) Packaging (see 5.1).
(e) Packing (see 5.2).
(f) Marking (see 5.4).

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MILITARY INTEREST:

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Army - ME

Review Interest:

CS

Preparing Activity:

GSA-FSS

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